

PROBLEM 1. TRANSFORMERS AS DEPENDENCY PARSERS

Part (a).*Proof.*

$$\alpha(\mathbf{dobj}, \mathbf{prefer}, w) = \delta_{w, \text{flight}}$$

□

Part (b).

Proof. We are interested in penalizing attention vectors that spread attention across multiple words, and rewarding attention vectors that focus on a single word. A natural candidate for this is the shannon entropy of the attention vector. Let p be a probability vector. Then

$$H(p) := - \sum_i p_i \log p_i$$

Since we wish to penalize all heads, we have the following $\mathcal{L}_{\text{focus}}$ definition:

$$\mathcal{L}_{\text{focus}} := \sum_{\ell, k, t_1} H(\alpha[\ell, k, t_1, T_2])$$

Additionally, we add a hyperparameter λ to control the strength of the regularization.

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda \mathcal{L}_{\text{focus}}$$

□

Part (c).

Proof. A natural candidate for penalizing distance from t_1 is just the absolute distance from t_1 , scaled by the attention weight. Squared distance is probably a better choice. Additionally, in order to cap the maximum penalty, to allow for the possibility of outliers, we introduce a hyperparameter L_{max} . As before, we introduce a hyperparameter λ to control the strength of the regularization.

$$\mathcal{L}_{\text{near}} := \sum_{\ell, k, t_1} \mathbb{E}^{\alpha[\ell, k, t_1, T_2]} [(t_1 - t_2)^2 \wedge L_{\text{max}}]$$

□

Part (d). The universality assumption is that the model will learn the optimal parameters for the language modeling task, so loss shaping will change the optimal parameters. In the real world, it is unlikely that the model will learn the optimal parameters, so loss shaping helps guide the model towards better parameters.

PROBLEM 2. ADJUSTING TEMPERATURE FOR DIMENSION

Consider adding a temperature parameter to the softmax:

$$\alpha[\ell, t, w] = \text{softmax}_w \beta h[\ell, t, I] e[w, I]$$

Part (a).*Proof.*

$$\begin{aligned} \mathbb{E}[\beta h[t, I] e[w, I]] &= 0 \\ \text{Var}(\beta h[t, I] e[w, I]) &= \beta^2 \text{Var}(h[t, I] e[w, I]) = \beta^2 \sum_i \mathbb{E}[h[t, i]^2] \mathbb{E}[e[w, i]^2] = \beta^2 I \end{aligned}$$

Therefore we let $\beta = I^{-1/2}$.

□

Part (b).

Proof. This is exactly the normalization used in the softmax.

□

PARAMETERIZING INNER-PRODUCT

Part (a). We require that we have multiple heads, so we need k to be a parameter.

Part (b).

$$P(y|T_2) = \operatorname{softmax}_{t_2} \frac{1}{\sqrt{I}} L[\ell, t_1, J]^\top W_{\ell+1}^Q[k, J, I] W_{\ell+1}^K[k, I, J] L[\ell, t_2, J]$$

PROBLEM 4. REPETITION AT LOW TEMPERATURES

Let $\beta = \infty$. Then we have:

$$\Pr(w_{t+2}|w_t, w_{t+1}) = \lim_{\beta \rightarrow \infty} \operatorname{softmax}_{w_{t+2}} \beta h[t+2, I] e[w_{t+2}, I]$$

which as seen in the first homework, is a one-hot distribution on the word with the highest inner product with $h[t+2, I]$. Each word is deterministic, so whenever we encounter a repeated pair of words, we will always repeat the same word, leading to infinite repetition.

PROBLEM 5. ELIMINATING THE KEY MATRIX

Self attention is computed by the following equations:

$$\begin{aligned} \text{Query}_{\ell+1}[k, t, i] &= W_{\ell+1}^Q[k, i, J] L_\ell[t, J] \\ \text{Key}_{\ell+1}[k, t, i] &= W_{\ell+1}^K[k, i, J] L_\ell[t, J] \\ \alpha_{\ell+1}[k, t_1, t_2] &= \operatorname{softmax}_{t_2} \left[\frac{1}{\sqrt{I}} \text{Query}_{\ell+1}[k, t_1, I] \text{Key}_{\ell+1}[k, t_2, I] \right] \end{aligned}$$

W^Q and W^K are of the same shape $[K, I, J]$. Typically $I < J$.

Part (a). Give an equation computing tensor $\tilde{W}^Q[K, J, J]$ from W^Q and W^K such that the attention can be written as:

$$\alpha_{\ell+1}(k, t_1, t_2) = \operatorname{softmax}_{t_2} \left[L_\ell[t_1, J_1] \tilde{W}^Q[k, J_1, J_2] L_\ell[t_2, J_2] \right]$$

Proof. We want:

$$\tilde{W}^Q[k, j_1, j_2] = \frac{1}{\sqrt{I}} W^Q[k, j_1, I] W^K[k, I, j_2]$$

Thus, we have

$$\tilde{W}^Q[k, J_1, J_2] = \frac{1}{\sqrt{I}} W^Q[k, J_1, I] W^K[k, I, J_2]$$

□

Part (b). What is the resulting number of query matrix parameters for a given layer, and how does this compare to the number of key-query matrix parameters for a given layer in the original transformer version.

Proof. We have K heads, and each head has J^2 parameters, so we have KJ^2 parameters. In the original, we have K heads, and each of the query and key matrices have IJ parameters, so we have $2KIJ$ parameters. If $J < 2I$, then we have fewer parameters than the original version, but if $I < \frac{J}{2}$, then we have more parameters than the original version. □